

EVAN J. LONG

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EDUCATION

The University of Texas at Austin – Austin, TX ----- August 2023 – Dec 2026
Bachelor of Science in Mechanical Engineering | GPA: 3.79/4.00 | 84 Course Hours
Relevant Coursework: Fluid Dynamics; Solids; Materials; Calculus; Thermodynamics; Statics; Physics; Chemistry; Programming; Engr. Projects

The University of New Mexico - Los Alamos – Los Alamos, NM ----- May 2021 – May 2023
Certificate in Robotics | GPA: 3.86/4.00 | 33 Course Hours
Relevant Coursework: Individual Robotics Projects; Manufacturing Techniques; Industrial Robotics; Drafting and CAD; Fluid, Cryo, Vacuum Systems.

EXPERIENCE

Texas Rocket Engineering Lab (TREL) – Austin, TX ----- January 2024 – Present
Test Engineer, Ground Fluids Responsible Engineer

- Led the ground fluids team in designing, building, and testing Ground Support Equipment (GSE) used for the “MKI” Halcyon Rocket.
- Refurbished and set up hydrocart, tubing, and blast shield to proof and autofrettage our Composite Overwrap Pressure Vessels (COPV’s)
- In overseeing the design, testing, and integration of fluid systems, prioritized safety, effectively managing risks associated with dangerous cryogenics and explosive oxidizers and fuels; notably, developed nearly the entirety of a FireX system by optimizing layout and pressure specifications to protect the engine bay and ensure clean integration with other hardware; Later cranked wrenches to build it.
- Facilitated effective cross-disciplinary teamwork and communication, accelerating the development timeline for critical systems, achieving key milestones for TREL’s ambitious goal of launching the first ever collegiate liquid-fueled rocket into space.

Maker/Mentor: College of Natural Sciences Makerspace – Austin, TX ----- September 2024 – Present

- Effectively my student peers and other visitors how to bring their ideas into reality, focusing on prototypes with 3d prints and laser cuts.
- Work from scratch on various projects to improve the space: A remote door opener using an ESP32 and my own design, An open-sourced plastic recycler with personalized optimizations (for prints), and hardware and procedure to make molds and cast into them.

ASCEND Paraglider Robot – Austin, TX ----- January 2025 – Present
Autonomous Steering RE, General Engineer

- Innovated a Capstan drive system to attach and turn a high-torque steering column while still using low-cost, low weight 3d printed parts.
- Created a model of the ATV chassis to ensure the assembly would fit perfectly in the allotted area

EXD Experimental Division Combat Robotics – Austin, TX ----- September 2024 – Present

- Have now built 3 different PLAntweight combat robots: One Hammersaw and two Full-Body Shell Spinners.
- Created intricate models and interfaces in OnShape with severe size restrictions, tested them repeatedly against dummy robots and other objects, and optimized the shape and material for resilience, impact, and weight.

Systems for Augmenting Human Mechanics (SAHM) Lab – Austin, TX ----- October 2023 – December 2024
Undergraduate Researcher, General Engineer

- Produced an orthosis to assist patients with impaired mobility (typically MLS) in walking by using TheraBands to redistribute work done by muscles in the rear of the leg to – degenerate – ones in the front, easing the swing phase and reducing likelihood of falls.
- Implemented an electromyography (EMG) system to analyze neuromuscular signals, converting activity into actionable data for visualization and general use in the lab, as well as controlling simulated and real-life prosthetics.

Nuclear and Applied Robotics Group (NRG) – Austin, TX ----- January 2024 – May 2024

- Developed, with two partners, a cutting-edge clay 3d printer module adaptable to any standard URD3 robotic arm, complete with a powered canister to store material, modular environment to allow for easier change implementation, and screw-driven extruder.
- Engineered this extruder from conception to function, addressing material and environment requirements to optimize performance.

Relevant Course Projects, UNM – Los Alamos, NM ----- January 2022 – May 2023

- Designed, built, tested, prototyped a functional robotic tank featuring a high-powered pneumatic cannon with movement in 2 directions.
- Repeated the process with an off-road vehicle focusing on a robust spring-based suspension, differential 4WD, and effective transmission
- Developed end effectors/environment for and programmed a FANUC standard industrial robot, enabling it to make, pour, and serve tea.

COMMUNITY INVOLVEMENT

Volunteer at Helping Hand Home (HHH) – Austin, TX ----- November 2024 – Present

- Came up with fun and engaging activities to get children excited about different STEM fields, mainly chemistry and engineering.
- Built relationships with kids from homes where they had been neglected/abused, and needed more than foster care could provide.

RELEVANT SKILLS

- **Modeling Software:** SolidWorks (including FEA), Ansys, OnShape, SketchUp, Fusion360 (CAM), FDM & SLA 3D Printing
- **Programming and Electronics:** Python, MATLAB; C++, ROS, General Electronics Layout and Assembly.
- **Certifications:** FANUC Industrial Robot Technician, Machine Shop, Programmable Logic Controllers, Safety Certifications (Fire, Cryogen, Compressed Gases, General Laboratory), Bilingual Seal for German Proficiency.

*LinkedIn alternate link: <https://www.linkedin.com/in/longevan217/>

**Portfolio website alternate link. View more of my projects here: <https://longevan217.wixsite.com/evan-long-portfolio>